
ECE228 Project Proposal: π_0 -Lite for Flow-Matching VLA Policy Evaluation

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Abstract

We propose π_0 -Lite, a course-scale reproduction of a key idea from recent robot foundation models: generating continuous robot action chunks using flow matching instead of discretized autoregressive action tokens. Given image observations, robot state, and a language instruction, the model predicts a future action sequence for language-conditioned manipulation. We will evaluate the method primarily on Isaac Lab-Arena environments through the Hugging Face LeRobot Environment Hub, with LIBERO or CALVIN as fallback offline benchmarks. The project compares flow-matching action generation against behavior cloning regression and discretized autoregressive action-token prediction, studying whether continuous generative action modeling improves action quality, smoothness, and inference efficiency for lightweight VLA policies.

1 Problem background and motivation

Generalist robot policies aim to map visual observations and language commands to robot actions across diverse manipulation tasks. Vision-language-action (VLA) models extend pretrained vision-language representations to robotic control, but a central challenge remains: robot actions are continuous, high-dimensional, and require smooth high-frequency execution, while many VLA systems generate actions as discrete tokens. Discretizing action dimensions allows the use of standard next-token prediction, but it can introduce quantization error, long output sequences, and slower inference for action chunks.

Recent π_0 proposes a different approach: augmenting a pretrained vision-language model with a flow-matching action expert that generates continuous action chunks. Reproducing the full π_0 system is beyond the scope of this course project because it relies on large-scale robot pretraining and substantial compute. Instead, we isolate one core design question: *Does flow matching provide a better lightweight action-generation mechanism than direct regression or autoregressive action tokens for language-conditioned robot manipulation?*

2 Related work

VLA models. OpenVLA is a 7B-parameter open-source VLA model trained on Open X-Embodiment data and adapted to downstream manipulation tasks [3]. Many VLA systems represent continuous actions as discrete tokens, making them compatible with language-model-style cross-entropy training, but this formulation may be less ideal for precise continuous control.

Generative action policies. Diffusion Policy models visuomotor control as conditional denoising over action sequences and shows strong performance on manipulation benchmarks [2]. π_0 extends

this idea to VLA policies by using flow matching to produce continuous high-frequency action chunks conditioned on visual and language inputs [1]. Our project studies this mechanism in a smaller, reproducible setting.

Benchmarks. Isaac Lab-Arena provides a scalable Isaac Lab-based framework for evaluating generalist robot policies in simulation, and its integration with LeRobot EnvHub enables environment registration and policy evaluation within the LeRobot ecosystem [6, 7]. LIBERO and CALVIN provide language-conditioned manipulation demonstrations and will be used as fallback benchmarks if full simulation integration is too time-consuming [4, 5].

3 High-level methodology

Given an image I_t , robot state s_t , and instruction l , the policy predicts an action chunk $A_t = [a_t, \dots, a_{t+H-1}] \in \mathbb{R}^{H \times d_a}$. At evaluation time, the policy is used in a receding-horizon manner: predict a chunk, execute the first action or first few actions, then replan.

π_0 -Lite model. The model contains: (1) a frozen or adapter-tuned visual encoder such as CLIP, ResNet, or a small ViT; (2) a pretrained text encoder such as CLIP text encoder or sentence-transformer; (3) an MLP state encoder; and (4) a conditional flow-matching action decoder. Image, language, and state embeddings are fused into a conditioning vector $c_t = f_\theta(I_t, s_t, l)$.

Flow matching objective. Let $x_1 = A_t$ be the expert action chunk and $x_0 \sim \mathcal{N}(0, I)$ be Gaussian noise with the same shape. For $\tau \sim \mathcal{U}(0, 1)$, construct $x_\tau = (1 - \tau)x_0 + \tau x_1$. The target velocity is $u_\tau = x_1 - x_0$, and the action decoder is trained with

$$\mathcal{L}_{\text{FM}} = \mathbb{E} \left[\|v_\phi(x_\tau, \tau, c_t) - (x_1 - x_0)\|_2^2 \right]. \quad (1)$$

During inference, the model samples an initial noisy action chunk and integrates the learned velocity field using a small number of Euler steps.

Baselines. We will compare against two baselines under the same observation setting. First, a behavior cloning regression model directly predicts continuous action chunks using MSE loss. Second, a discretized autoregressive action-token model converts each action dimension into bins and predicts the resulting token sequence with cross-entropy loss, then maps tokens back to continuous actions.

4 Evaluation plan and expected outcome

The primary evaluation target is Isaac Lab-Arena via LeRobot EnvHub. We will implement a policy wrapper that converts environment observations into the π_0 -Lite input format and converts predicted action chunks into environment actions. If Isaac Lab-Arena integration becomes too time-consuming, we will evaluate on LIBERO or CALVIN offline demonstrations. Metrics include: task success rate if rollout is available, action MSE/L1, action smoothness, average episode length, and inference latency. Planned ablations include action horizon $H \in \{4, 8, 16\}$, flow inference steps $K \in \{4, 8, 16\}$, and with-language versus without-language conditioning.

The expected contribution is not a full reproduction of π_0 , but a controlled study of its action-generation mechanism. The final deliverable will include: (1) a lightweight flow-matching VLA implementation, (2) comparison with regression and autoregressive-token baselines, and (3) empirical analysis of whether continuous flow-based action chunks improve lightweight language-conditioned manipulation.

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